

1750 – 2024

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven analysis from the Industrial Revolution to 2024

Global Carbon Budget 2025

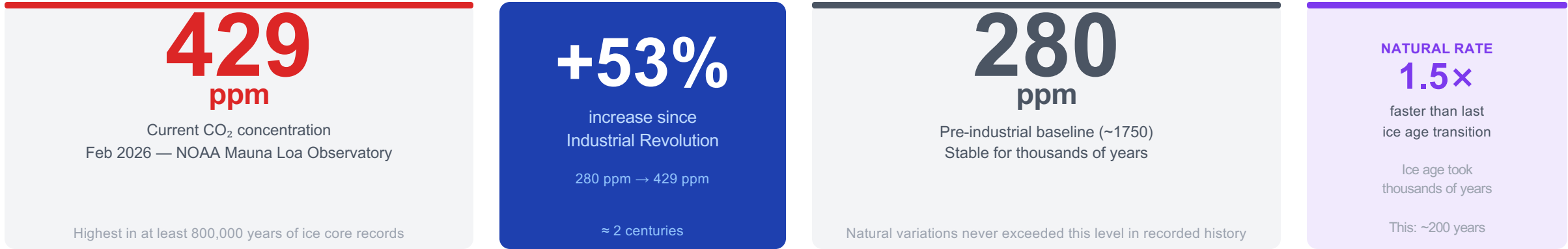
NASA / NOAA

Our World in Data

CO₂

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

At 429 ppm, atmospheric CO₂ is now more than 50% above pre-industrial levels — a change that took thousands of years naturally but has occurred in roughly two centuries.



Key Context

**Rate of Change Is Unprecedented**
At least 1.5× faster than the natural CO₂ increase observed at the end of the last ice age 20,000 years ago.

**Continuous Monitoring Since 1958**
NOAA's Mauna Loa Observatory has tracked atmospheric CO₂ since 1958 (the Keeling Curve). The trend shows no sign of reversal.

**Fossil Fuels Are the Primary Driver**
Ground-based and satellite measurements confirm the sharp acceleration is linked primarily to coal, oil, and natural gas.

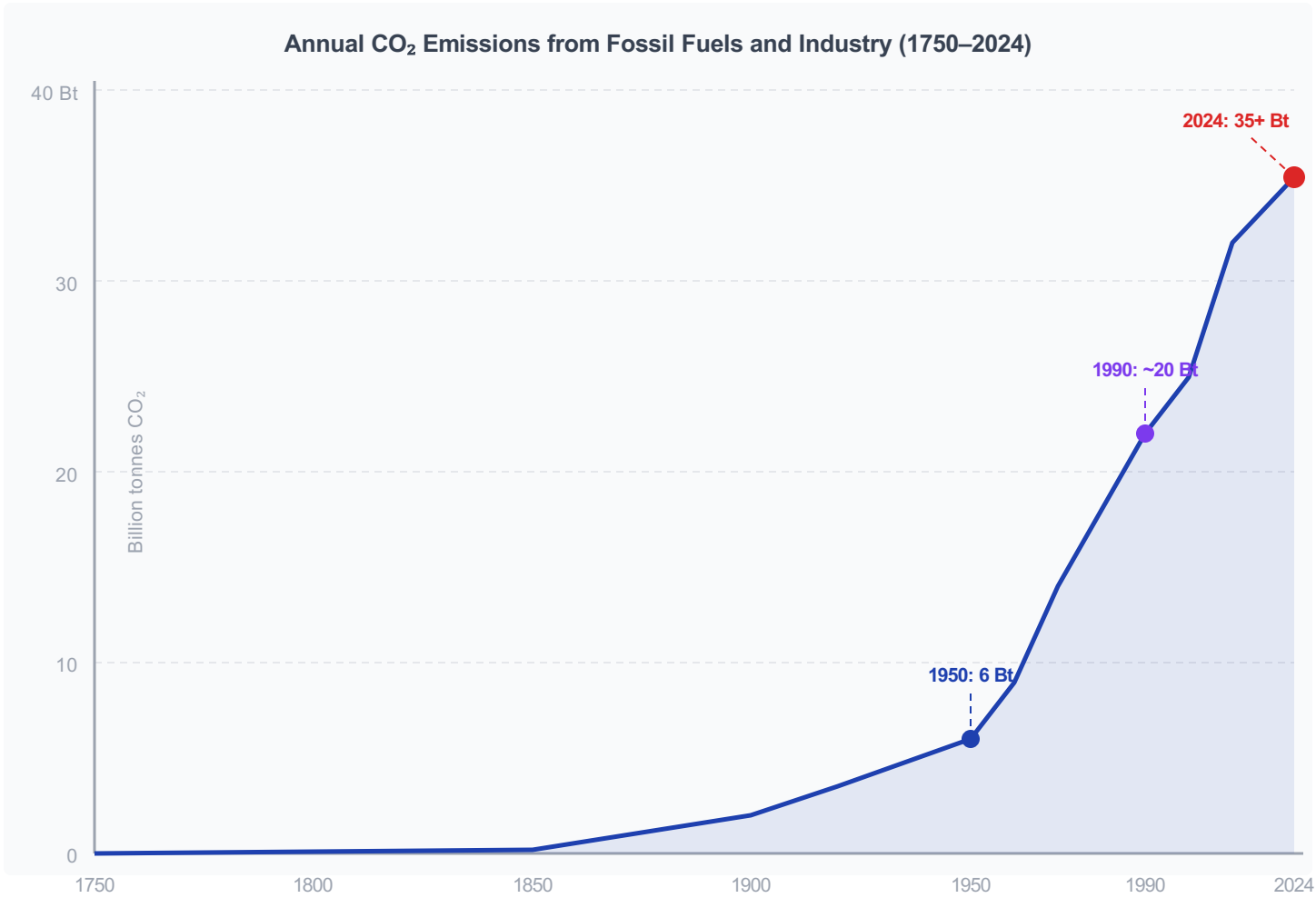
CO₂ TRAJECTORY OVERVIEW



Strategic Implication: Current CO₂ levels are irreversible on human timescales. Every year of delayed action increases the atmospheric burden permanently. This is the scientific backdrop for all current climate negotiations.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Global emissions have grown more than six-fold since 1950 — from 6 Bt to 35+ Bt today — and have not yet peaked. Growth was near-flat before 1900, then accelerated sharply through industrialization and the post-war boom.



6 Bt
1950 — Post-War Baseline
Global emissions as the modern industrial age accelerated. 6× growth was ahead.

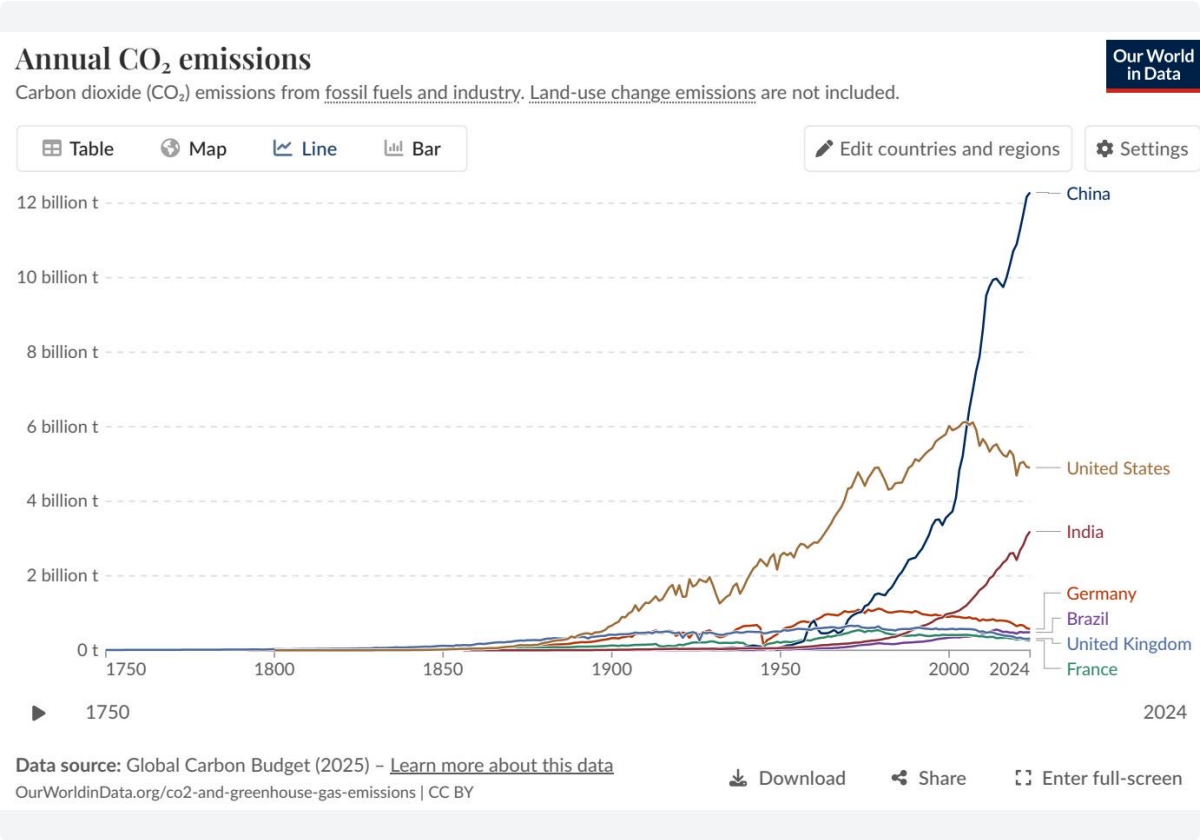
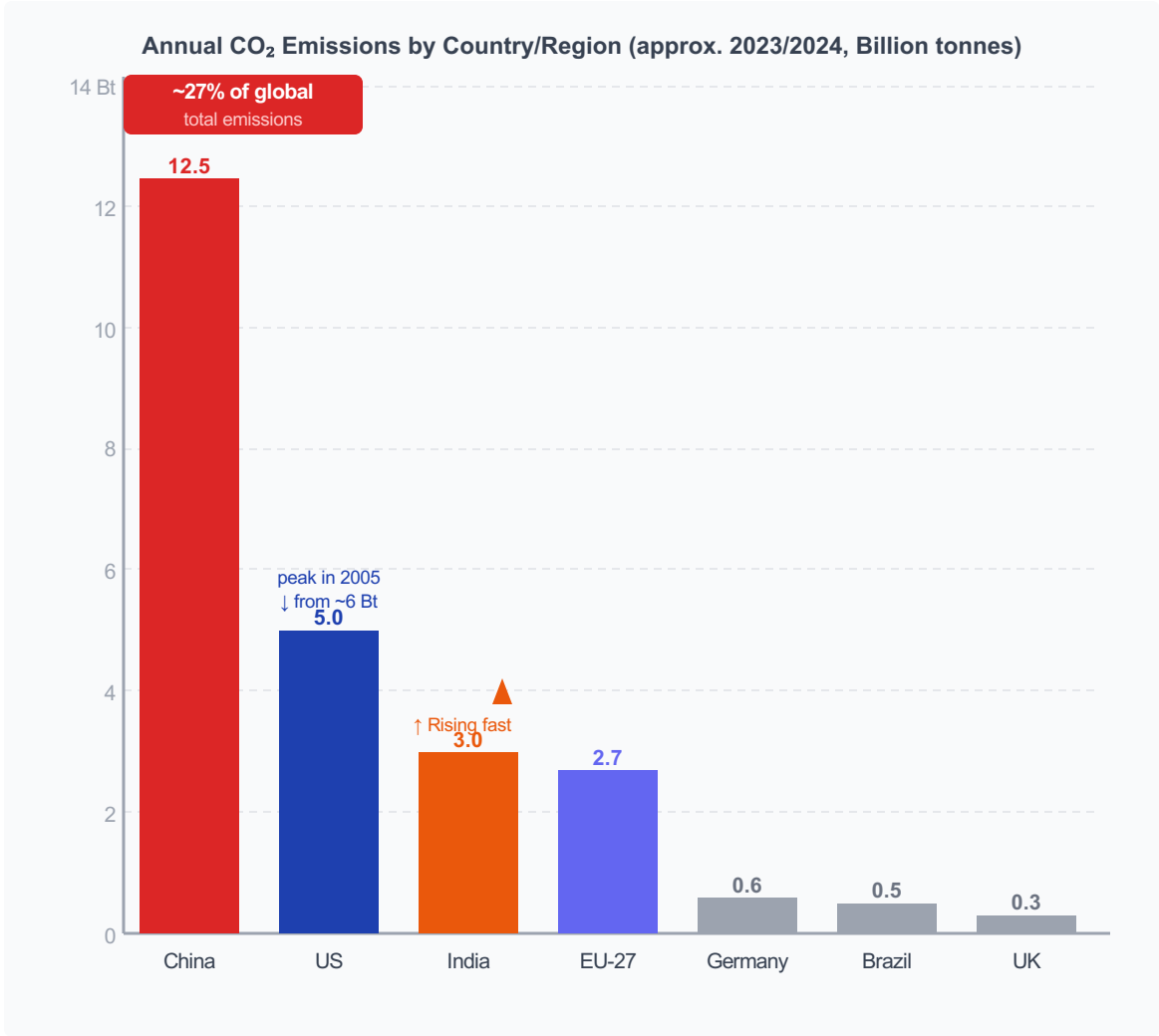
~20 Bt
1990 — UNFCCC Baseline Year
Nearly quadrupled from 1950. The Kyoto Protocol used 1990 as its reference year.

35+ Bt
2024 — Still Not Peaked
Growth slowing but absolute level rising. No year-on-year decline yet in global total.

6× growth in 74 years (1950–2024)
Despite efficiency gains and renewables expansion

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

China alone accounts for ~27% of global CO₂ emissions — more than double the US (5 Bt). India (~3 Bt) is the third-largest emitter, surpassing the EU-27, and is still growing rapidly.

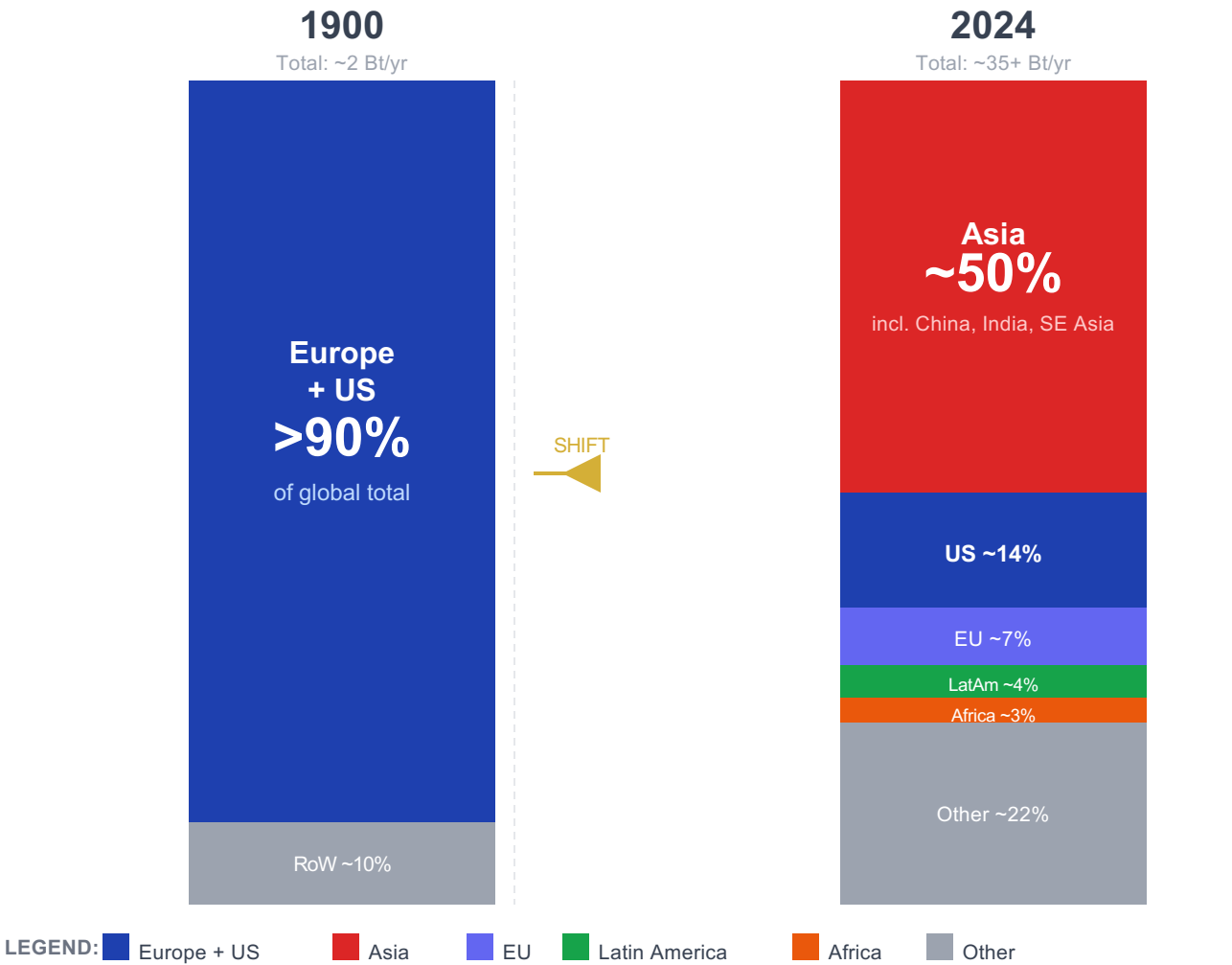


Annual CO₂ emissions by country, 1750–2024 (Global Carbon Budget 2025)

Country	Bt/yr	Global %	Trend
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising ↑

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

The geographic center of emissions has fundamentally shifted: from Europe + North America (>90% in 1900) to Asia (~50% today). Africa and South America each contribute only 3–4% — comparable to international aviation and shipping.



~50% Asia (2024)
China (~27%), India (~8%), and the rest of Asia now produce roughly half of all global CO₂ emissions.
vs. <<10% in 1900

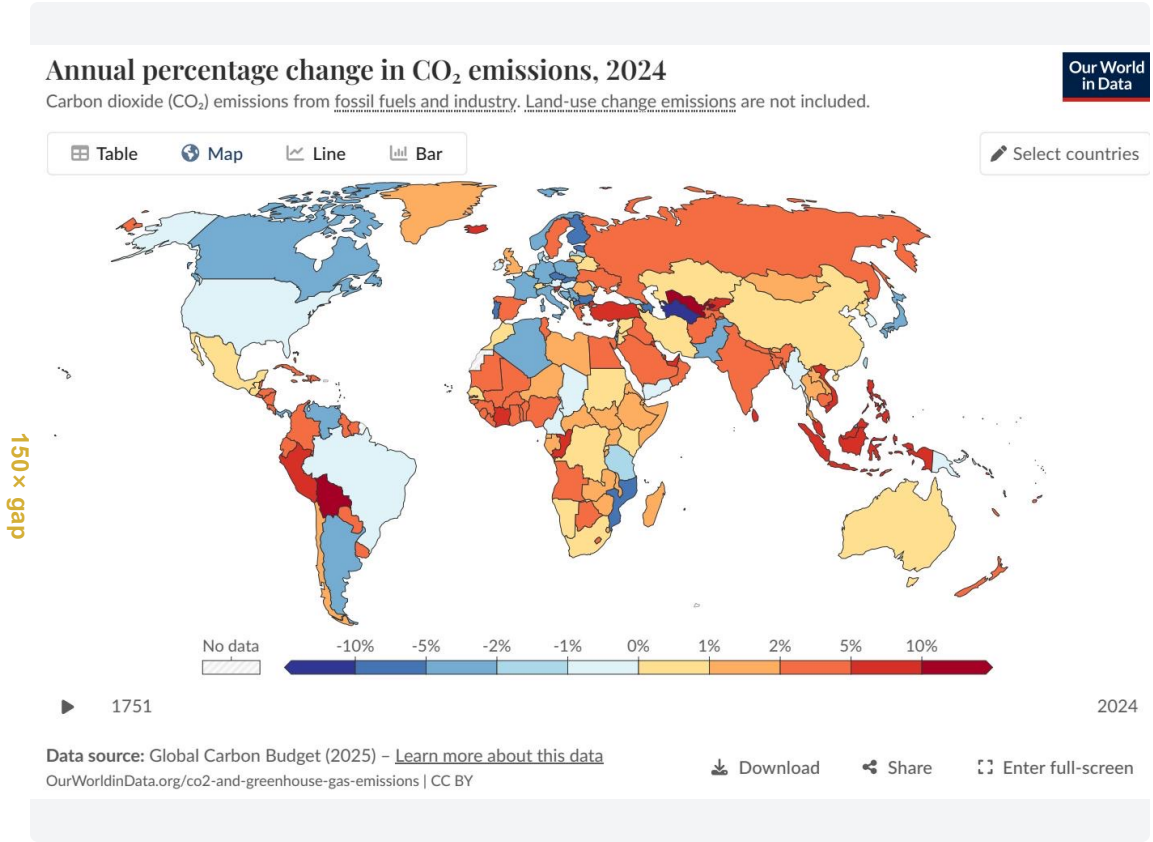
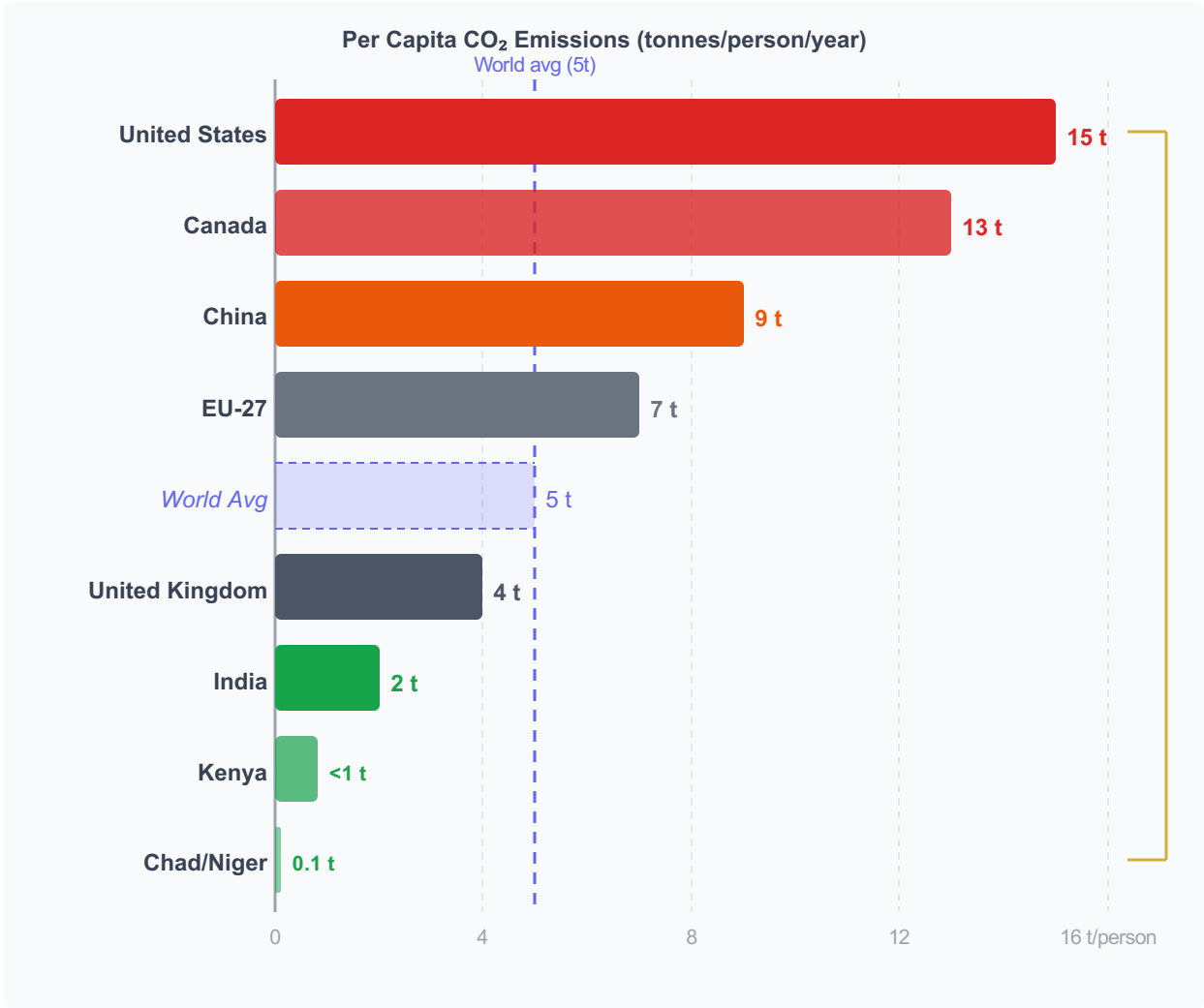
<33% US + Europe (2024)
US (~14%) + EU (~7%) = less than one-third of global emissions. Still declining, but slowly.
vs. >90% combined in 1900

3–4% Africa + South America Each
Each region contributes just 3–4% of global emissions — comparable to international aviation and shipping combined.

Population Context:
Asia is home to ~60% of the world's population yet produces ~50% of emissions — closer to proportional parity than the historical 1900 picture where Europe+US used 90% on 25% of people.

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

The average American emits in under two days what the average person in Chad emits in an entire year — a 150-fold disparity that defines the equity dimension of climate negotiations.



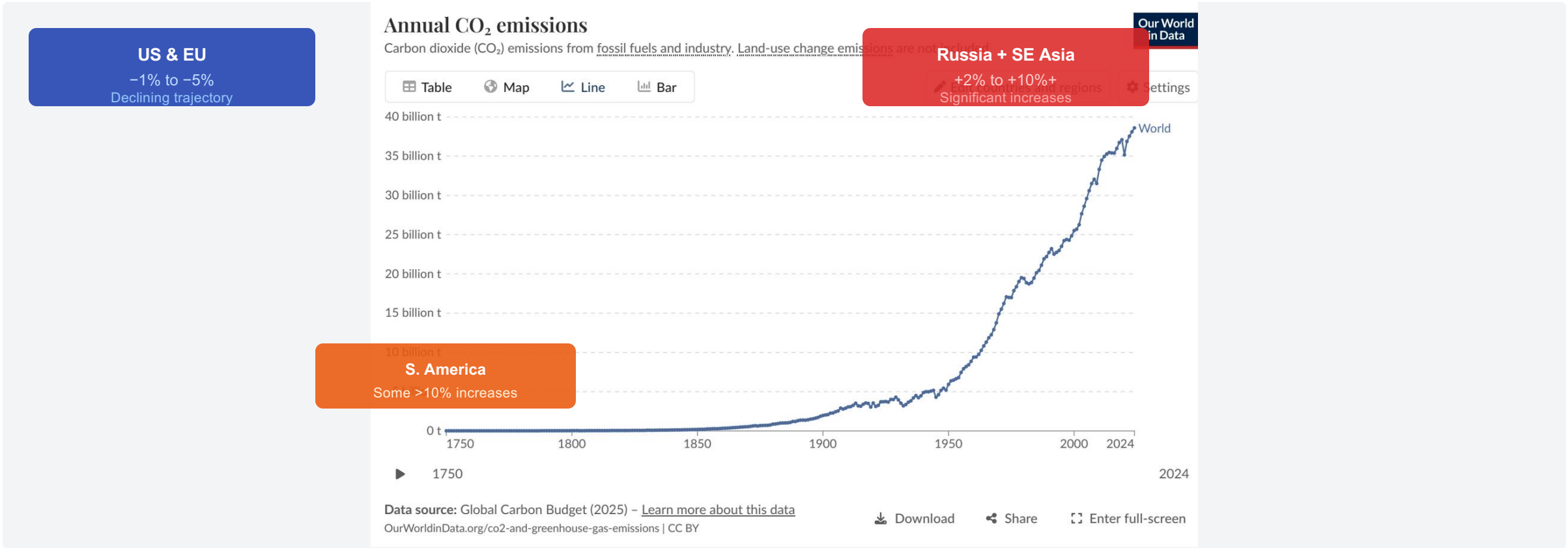
Per capita CO₂ emissions over time, selected countries (Our World in Data)

Key Equity Insight:

Countries with similar prosperity can differ greatly in per capita emissions. France (~4 t) vs. Germany (~7 t): energy mix — not GDP — explains the gap.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

The 2024 emissions map shows a diverging world: the US and most of Europe are cutting (−1% to −5%), while Russia, Southeast Asia, and parts of Africa are increasing (+2% to +10% or more).



Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries at similar prosperity levels can have dramatically different per capita emissions. Energy mix — especially the share of nuclear and renewables vs. fossil fuels — is the key differentiator, not GDP alone.

France & United Kingdom

~4 t

per person per year

- High nuclear share (France: ~70% of electricity)
- Growing renewable energy portfolio
- Relatively low coal and gas dependence
- Similar GDP per capita to Germany/Netherlands

Below world average (5 t)

VS

Germany & Netherlands

~7–8 t

per person per year

- Germany phased out nuclear (post-Fukushima)
- Higher reliance on natural gas and coal
- Netherlands: significant gas infrastructure
- Renewables growing but fossil still dominant

Nearly double France/UK per capita

Energy Mix & Per Capita CO ₂ Comparison				
Country	CO ₂ /person	Nuclear	Renewable	Fossil Reliance
France ~0.8% global share	~4 t	High	Moderate	Low
United Kingdom ~0.9% global share	~4 t	Moderate	High	Declining
Germany ~1.7% global share	~7 t	Phased out	Growing	High (gas+coal)
Netherlands Small country, high per capita	~8 t	Low	Growing	High (gas)
Policy Implication for Negotiators: Equity-adjusted emission frameworks cannot rely on GDP alone. Countries at similar development levels show 2× per capita differences driven by energy policy choices.				
Germany emits ~75% more per person than France despite comparable economic development — entirely explained by energy mix				

Key Takeaways: Emissions Still Rising, But the Map of Responsibility Is Shifting

Five facts every climate negotiator needs: emissions are still rising, but who is responsible — historically and per capita — is more nuanced than total tonnage alone.

1 35+ Billion Tonnes and Still Climbing

Global annual CO₂ emissions have not yet peaked. Growth is slowing but absolute levels continue to rise, driven by Asia and other regions.

Source: Global Carbon Budget 2025

2 China Leads at 27%, India Rising Fast

China emits ~12.5 Bt (~27% of global total). India (~3 Bt, ~8%) is the third-largest emitter, growing rapidly, and has surpassed the entire EU-27.

Source: Global Carbon Budget 2025

3 150× Per Capita Inequality

US: ~15 t/person vs. Chad: ~0.1 t. The average American emits in under two days what the average Chadian emits in an entire year.

Source: Global Carbon Budget 2025

4 US and Europe Are Declining, But Historical Responsibility Remains

US emissions are down ~-1% to -5% in 2024; EU similarly. But cumulative historical emissions from industrialized nations remain large — a key point in loss-and-damage and finance negotiations.

Source: Global Carbon Budget 2025

5 Energy Mix Is the Critical Policy Lever

France (~4 t) vs. Germany (~7 t): comparable GDP, dramatically different emissions. Nuclear and renewables — not prosperity alone — determine per capita outcomes.

Source: Global Carbon Budget 2025

Overall: Global emissions must peak and fall sharply within this decade. The path forward requires addressing both total national tonnage (China, India) and per capita equity (US, Australia) — alongside accelerated clean energy transitions.